

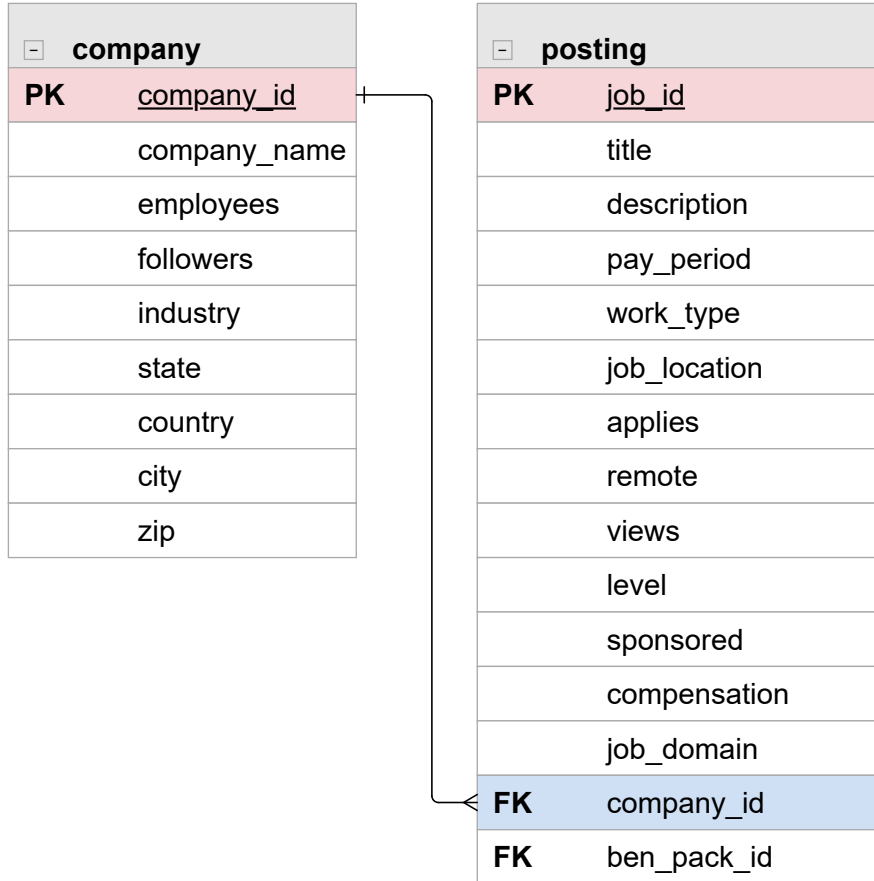
IS 201 TA Lab



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ERD: Primary / Foreign Keys



A **primary key** is a unique identifier (think social security number)

A **foreign key** is another table's primary key, used to link two tables together

Entity
Relationship
Diagram

ERD: Cardinality

Read left-to-right AND right-to-left

One to One (1:1)

A store has one manager

A manager works at one store

One to Many (1:N)

A customer can have multiple orders

An order belongs to one customer

Many to Many (M:N)

A student can enroll in many courses

A course can have lots of students

Note! M:N cardinality requires a composite table (next slide)

ERD: Composite Table

movie	
PK	<u>movieid</u>
	title
	mpaa_rating
	budget
	gross
	release_date
	genre
	runtime
	rating
	rating_count

character	
PK/FK	<u>movieid</u>
PK/FK	<u>actorid</u>
	character_name
	credit_order
	pay
	screentime

actor	
PK	<u>actorid</u>
	name
	date_of_birth
	birth_city
	birth_country
	height_inches
	biography
	gender
	ethnicity
	networth

To know how much money Tom Hanks got **paid** to play Woody in *Toy Story*, you need to know both the **movie** and the **actor**. His pay is probably different than when he was Woody in *Toy Story 2*, or when he was Forrest Gump in *Forrest Gump*

ERD: Minimum / Maximum Cardinality

The **inner** ones are **minimum** (0 or 1)

and the **outer** ones are **maximum** (1 or many)

Table A	Table B
1 B or many B	1 A or 1 A
0 B or many B	1 A or many A
1 B or 1 B	0 A or 1 A
0 B or 1 B	0 A or many A

Think: a student could have 0 cars, but they could also have multiple.

SQL

Structured Query Language



```
1  SELECT (DISTINCT)  -- attributes
2  FROM               -- tableA
3  JOIN               -- tableB ON tableA.attribute = tableB.attribute (order doesn't matter)
4  WHERE              -- attribute filters [=, !=, <>, IS, LIKE '%__%', IN ("__","__","__")]
5      AND/OR         -- you only say WHERE once, but can have many filters
6  GROUP BY           -- all non-aggregated attributes when aggregating
7  HAVING              -- group-based filters (not used in the SQL project)
8  ORDER BY           -- attributes in ASC (default) / DESC order
9  LIMIT              -- to xx top results
10
11 -- aggregate(attribute) AS newName
12 count()
13 avg()
14 min()
15 max()
16 etc.
```

SQL: Where Use Cases

ta_name = "James"

→ Exact match

ta_name != "James"

→ Not exact match

ta_name <> "James"

(these are equivalent)

ta_name IS NULL

→ NULL means blank data (can also do **IS NOT NULL**)

ta_name LIKE '%ame%'

→ The text ("ame") is contained within the attribute

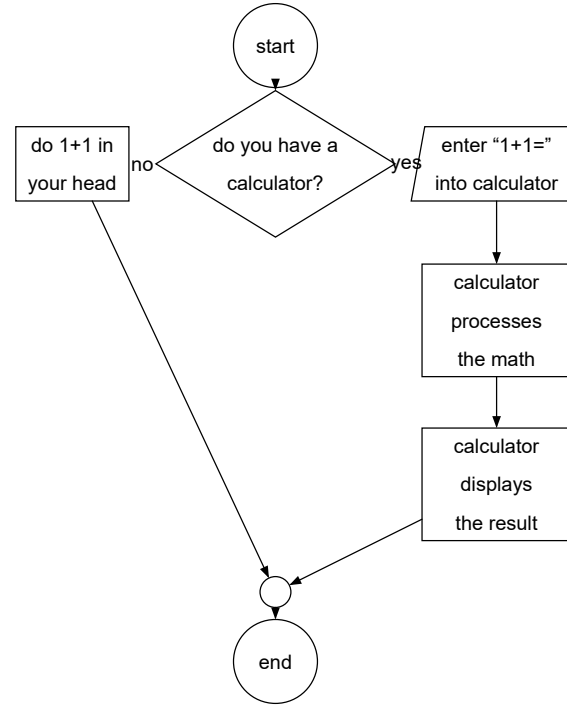
(% = wildcard)

ta_name IN ("James", "Robert", "Frankie")

→ Exact match for any of these

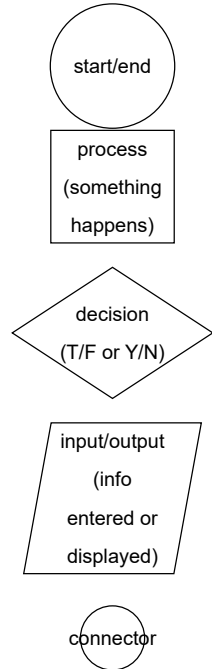
Flow Charts

EXAMPLE: How to solve 1+1



go to

draw.io



you can only have ONE start, but you CAN have multiple ends.

decisions are the ONLY shape with >1 arrow OUT.

connectors are the ONLY shape with >1 arrow IN.

file > export as > pdf

```
1 Option Explicit 'This makes it so that you can only use variables you've declared (VERY RECOMMENDED)
2 Sub thisIsMySubName()
3     'BASICS!
4     'Generally you will first refer to an object type (like a Sheet)
5     'and then tell it what you want to do (like delete or add or copy, etc):
6     ' Sheets and Worksheets are the same thing
7     Sheets("MySheet").Delete
8     Sheets.Add.name = "MySheet"
9     Worksheets("MySheet").Activate
10    ' Range and Cells are SIMILAR but different: Range("E7") = Cells(7, 5)
11    Range("A1").Activate
12    ActiveCell.Value = "Hello"
13    Range("A1:D3").Copy
14    Range("E5").PasteSpecial
15    ' Columns and Rows are what they sound like
16    Columns("B:D").Delete
```

VBA: Navigation and Misc.

Visual Basic

```
1 'NAVIGATION!
2 '
3     ActiveCell.End (xlDown)
4     ActiveCell.End (xlToLeft)
5
6     ' Move (rows, columns) based on a cell or range
7     ActiveCell.Offset(1, 2).Value = "NewSpot"
8         Range("A1").Offset(3, 4).Value = "NewSpot" 'The cell that now says "NewSpot" is E4
9
10 ' MISCELLANEOUS!
11
12     ' Bolding
13     Range("A1").Font.Bold = True
14
15     ' Autofitting columns
16     Columns.AutoFit 'all columns
17     Columns("A:C").AutoFit 'specific columns
18
19     ' Currency format
20     Range("A1").Style = "Currency"
21
22     ' Bottom border
23     Range("A1").Borders(xlEdgeBottom).LineStyle = xlContinuous
24
25     ' Text splicing
26     name = "James"
27     first_initial = Left(name, 1)
```

VBA: Variables

VB Visual Basic

```
1 'DECLARE VARIABLES!
2     Dim i As Integer    'a whole number
3     Dim j As Double     'a number with a decimal
4     Dim k As String     'a line of text
5     Dim l As Boolean    'true or false
6     Dim m As Worksheet
7
8     i = 5
9     j = 1.2345
10    k = "Hello there! General Kenobi!"
11    l = True
12    Set m = Sheets("Sheet1") 'objects (sheets, books, ranges) use Set instead of =
13
14 'USER INTERFACE!
15
16     'Tell the user something
17     MsgBox "Oh hello there user, welcome to my spreadsheet"
18
19     'Ask for an input
20     Dim username As String
21     username = InputBox("What's your name?")
22
23 ' COMBINGING STRINGS
24
25     Dim myOutput As String
26     Dim name As String
27     name = "James"
28
29     ' COmbine strings with ampersands (&)-- don't forget spaces
30     myOutput = "Hello. My name is " & name & " and this is Disney Channel!"
```

VBA: Conditionals

VB Visual Basic

```
1  'CONDITIONALS!
2      ' If Statements
3      If i < 0 Then
4          ' your code here
5      ElseIf i < 5 Then
6          ' your code here
7      ElseIf i < 10 Then
8          ' your code here
9      Else
10         ' your code here
11      End If
12
13      'Case Statements
14      Select Case fruit
15          Case "apple", "strawberry"
16              Color = "red"
17          Case "grape"
18              Color = "purple"
19          Case Else
20              Color = "unknown"
21      End Select
22
23      'Another Case Statement
24      Select Case temperature
25          Case Is >= 100
26              feel = "too hot"
27          Case 80 To 100
28              feel = "pretty warm"
29          Case Is < 80
30              feel = "I'm probably staying out of the pool"
31      End Select
```

VBA: Loops

VB Visual Basic

```
1 'LOOPS!
2 'For Loop
3 For i = 0 To 5
4     ' your code here
5 Next i
6
7 'For Each Loop
8 Dim x As Range
9 For Each x In Range("A2", Cells(lastrow, "A"))
10     ' your code here
11 Next
12
13 'For Each Loop 2
14 Dim x As Range
15 Dim mtRange As Range
16 Set myRange = Range("A1:E5")
17 For Each x In myRange
18     ' your code here
19 Next
20
21 'Do While Loop
22 Do While i = 5 'condition does NOT have to be numbers
23     ' your code here
24 Loop
25
26 ' Do Until Loop
27 Do Until i = 10 'condition does NOT have to be numbers
28     'your code here
29 Loop
30
31 End Sub
```

VBA: Functions

VB Visual Basic

```
1 Function thisIsMyFunctionName(num As Integer, tf As Boolean) As String
2
3     ' a function declaration is just like a set of dims
4     ' the INPUT is in the parentheses ()
5     ' the OUTPUT is outside of it
6     ' so this function KIND OF has this baked into it:
7
8     Dim num As Integer
9     Dim tf As Boolean
10    Dim thisIsMyFunctionName As String
11
12    ' the INPUT then comes from your excel sheet
13    ' for example, in your sheet you cell might say:
14    ' =thisIsMyFunctionName(15, True)
15
16    ' the OUTPUT comes from your function
17    ' set the function's name equal to something
18    ' in this case, our output has to be a string:
19
20    thisIsMyFunctionName = "That's a very nice hat you have there, partner!"
21
22    ' generally your inputs will be used to calculate/decide your output
23
24 End Function
```

Null Hypothesis

H₀: No x's are significant

Alt Hypothesis

H_A: At least one x is significant

Scientific Notation:

$aEb = a \times 10^b$

$aE3 = a \times 1000$ (big)

$aE(-3) = a \times 0.001$ (small)

Statistics

Test	Variables	Stat
Correlation	numeric – numeric	r
Regression	numeric – multi. numeric	R^2
ANOVA	numeric – mult. categorical	p
T-test	numeric – 2 pair categorical	p

r : -1 to 1 (as $x \uparrow$, $y \downarrow$ / $y \uparrow$)

R^2 : 0 to 1 (share of variance in y explained by x)

p : 0 to α (usually 0.05) $\rightarrow$ reject the null if $p < \alpha$

Tableau

- “Default” settings are usually a good starting point — check them first
- Switching the x- and y-axis can reveal a clearer story
- Watch your sort/ordering — Tableau doesn’t always order the way you expect

Solver

Two important Excel formulas:

```
1 =SUM(A1:A5)           → A1 + A2 + A3 + A4 + A5
2 =SUMPRODUCT(A1:A5, B11:B15) → A1*B11 + A2*B12 + ... + A5*B15
3                        (the two ranges should be the same size and shape)
```

- **Constraints:** set a changing-cell reference, an operator ($\leq$, $=$, $\geq$), and the constraint value
- **If integer constraints aren't working:** Solver → Options → uncheck “Ignore Integer Constraints” and set Integer Optimality to 1%

HTML: Setup

HyperText Markup Language

In VS Code, inside a .html file, type ! then Enter to generate:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Document</title>
7  </head>
8  <body>
9  </body>
10 </html>
```

HTML

startbootstrap.com/themes/portfolio-resume

HTML: Tags



```
1  <!-- This is a comment -->
2  <h1>Main heading</h1>
3  <h2>Section heading</h2>
4  <h3>Subsection</h3>
5  <p>This is a paragraph.</p>
6  <!-- Line break -->
7  <br>
8  <!-- Unordered list (bullets) -->
9  <ul>
10     <li>Item 1</li>
11     <li>Item 2</li>
12 </ul>
13 <!-- Ordered list (numbers) -->
14 <ol>
15     <li>Item 1</li>
16     <li>Item 2</li>
17 </ol>
```

The start and end tags act like parentheses: <tag> stuff </tag>

HTML: Anchors

```
1 <!-- Hyperlinks -->
2 <a href="https://www.google.com">Visit Google</a>
3 <!-- Folder references -->
4 <a href="otherpage.html">Visit Other Page</a>
5 <a href="subfolder/page.html">Visit Subfolder Page</a>
6 <a href="../yetanotherpage.html">Visit yet another page</a>
7 <!-- On-page anchors (any tag can have an id, not just <a>) -->
8 <a id="section1">Section 1</a>
9 <a href="#section1">Go to Section 1</a>
```

HTML

../ means go to the parent folder (or root directory)

HTML: Images

```
1 <!-- Images -->
2 
3 
```



Notice: images do NOT have an end tag `</img>` (neither do line breaks `</br>`)

HTML: Divisions

Divs don't inherently do anything — they're invisible boxes you put content into so you can isolate it for positioning and CSS styling using classes.

HTML

```
1 <div class="square-image">
2   
3 </div>
```



CSS

```
1 .square-image {
2   width: 200px;
3   height: 200px;
4 }
```



For more, see: [CSS: Selectors/Properties/Values](#)

Sometimes doesn't work until you go live

YouTube

1. Click **Share**
2. Click **Embed**
3. Click **Copy**
4. Paste into your code

HTML: Embedding

Tableau

1. Click the share icon
2. Click **Copy Embed Code**
3. Paste into your code

HTML: Folder Structure & Formatting

Use folders! Keep `css/`, `html/`, `images/`, and `js/` separate, with `index.html` at the project root.

To quickly format everything correctly (tabs, long lines, etc.) in VS Code:

Right click → Format Document

HTML: Uploading to GitHub

1. Click **New** to create a repository
2. Click **Create repository**
3. On the repo page: **Add file** → **Upload files**
4. Click **Commit changes**

Upload the CONTENTS of your folder, not the folder itself

Going Live

1. Go to **Settings**
2. Click **Pages** in the sidebar
3. Under Branch, select `main` (instead of `None`) and **Save**

After a couple of minutes, refresh the page — there will be a link to your website at the top.

CSS: Basics

Cascading Style Sheets

```
1 <!-- Connect CSS to HTML, in your <head> -->
2 <link rel="stylesheet" href="styles.css" />
```

HTML

```
1 selector {
2     property: value;
3     property2: value2;
4 }
```

CSS

CSS is **cascading** because more recent styling supersedes previous styling — if you set the font color to blue, then later set it to red, it will be red.

CSS: Selectors / Properties / Values

```
1  /* Built-in selectors: the tags you use in your HTML (body, h1, p, a, img...) */
2  body {
3      background-color: #f0f0f0;
4      font-family: Arial, sans-serif;
5      margin: 0;
6      padding: 20px;
7  }
8  /* Class (custom) selectors: create your own with a period + name */
9  .highlight {
10     background-color: yellow;
11     font-weight: bold;
12 }
```

CSS

```
1  <p class="highlight">This text will be highlighted.</p>
```

HTML

There are A LOT of selectors and properties — Google and AI are your friend for finding exactly what you want!

jimna-h.github.io/IS_201_CHEATSHEET

